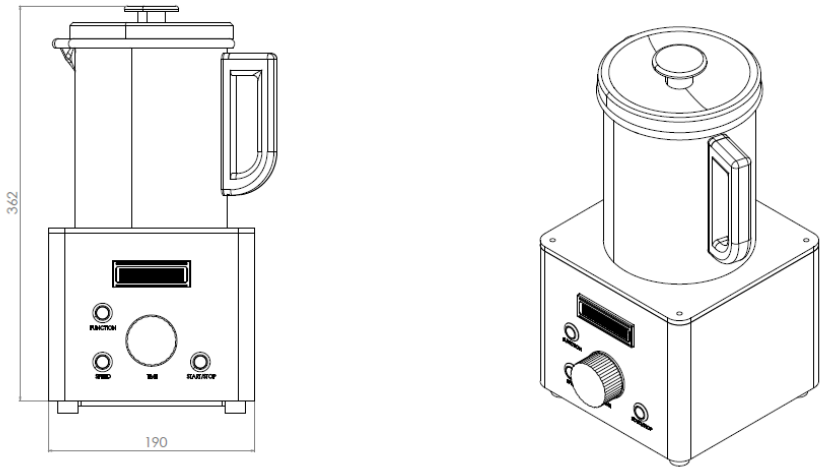
PROTOTYPE EDITION 1.0

# Nut Milk Maker

User Guide | November 2026




*Product concept: original team CAD drawing*

An open-source, modular countertop appliance that grinds, mixes and heats plant-based drinks. It is built from three replaceable modules: the jug (bowl), the blade and the motor.

**PROTOTYPE NOTICE**

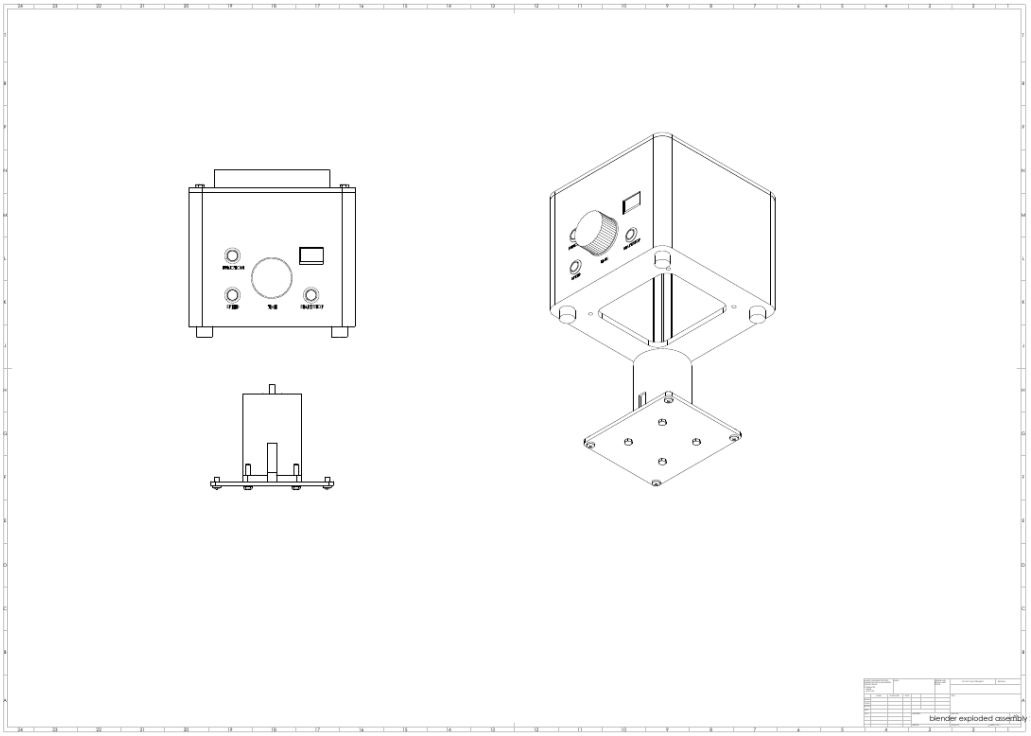
This guide describes a prototype. It is **not a certified consumer product.** Please use the machine only after the team has finished the electrical, heat, food-contact and lid-switch safety tests.

## Contents

**1** Important safety information
**2** Before first use
**3** Controls and programs
**4** How to make nut milk
**5** Cleaning and care
**6** Assembly and parts
**7** Troubleshooting
**8** Technical specifications
**9** Repair, circular design and open-source license
**10** Self-test and workaround modes

### The machine at a glance

- **Jug:** a removable jug with a handle and the blade and shaft assembly.
- **Motor base:** holds the motor, the front controls and the display.
- **Heater, temperature sensor and lid safety switch:** these are planned functions. Please check each one on your own prototype before relying on it.




*Motor base and front control layout: team engineering drawing*

## 1. Important safety information

Please read this section before you plug the machine in. It keeps you, the people around you and the machine safe.

**Where to use it**

- Use it on a **level, dry, heat-resistant surface** with clear space around it for ventilation.
- Keep it away from children.

**Water and electricity**

- **Never put the motor base, cable, plug or control electronics in water.**
- **Unplug the machine** before cleaning, servicing or changing any module.
- The 210–240 V AC mains circuit must be built and checked by a **qualified person**. Never use a prototype with exposed mains wiring.

**Blades and lid**

- The blades are sharp and can keep spinning for a moment after the machine stops. Never reach into the jug while the machine is powered.
- Do not open the lid or lift the jug while the motor or heater is running. The lid/jug safety switch must stop the machine whenever the lid or jug is not in place.
- **Never bypass a safety switch.**

**Heat and steam**

- Hot liquid and steam can scald. Let the machine cool before you open it or pour.
- Never close a hot jug so tightly that pressure can build up.

**Filling and running**

- Never run the machine dry, and never fill above the verified maximum mark.
- **Stop at once** if you see a leak, strong vibration or smoke, or notice a strange smell or noise.

**Parts**

- Do not use parts that are cracked, loose, bent, damaged or not food-safe.

> **NOT YET PROVEN**
> The safety limits, food-contact materials, insulation class and certifications have not been demonstrated in the project documents yet. They must be checked and confirmed before anyone outside the team uses the machine.

## 2. Before first use

Go through these five steps before the first run.

1. **Check for damage.** Look over the jug, handle, blade fastener, seal, motor coupling, lid, cable and housing.
2. **Wash the removable food-contact parts** and rinse them well. Keep every electrical part dry.
3. **Fit the jug and lid.** Make sure the blade turns freely while the machine is unplugged, and check that nothing leaks.
4. **Test the safety switch.** The project team does this with a safe, unloaded commissioning test. Please do not skip it.
5. **Match the power to the motor.** Check that the supply and motor driver suit the motor that is actually fitted. The baseline is a 12 V DC motor, and the final voltage is still to be confirmed.

## 3. Controls and programs

The machine is run by an ATmega328P (Arduino Nano V3) controller. The table follows the firmware, so please check the labels and positions against your final control panel.

| Control | What it does |
|---|---|
| **START / STOP** | The most important control. Press once to start. Press again to switch off the motor and heater immediately. |
| **TIME knob** | Sets the run time in manual mode: 30 to 60 minutes, in 5-minute steps. |
| **PROGRAM button** | Each press moves to the next program: Manual, Recipe 1, Recipe 2, then Eco-Self Clean, and back to Manual. |
| **SPEED button** | Sets the motor speed from 50% to 100%, in 10% steps. |
| **LCD 1602 display** | Shows the status and your settings. |
| **Lid / jug switch** | A safety input. The machine only runs when the lid and jug are closed and seated correctly. |

### Preset settings

| Program | Good for | Time | Speed |
|---|---|---|---|
| **Manual** | Almonds, cashews, hazelnuts | 30 mins (default) | 50% (default) |
| **Recipe 1** | Cashews, macadamia nuts, pine nuts | 45 mins | 70% |
| **Recipe 2** | Almonds, walnuts, hazelnuts | 50 mins | 60% |
| **Self-Clean** | Use for self cleaning | 10 mins | 25% |

While a program runs, the motor works in pulses: **60 seconds on, then 30 seconds of rest.** When the cycle ends, the buzzer sounds for 5 seconds and all outputs switch off.

> **GOOD TO KNOW**
> These are the design settings. The firmware has not been fully validated yet, so treat them as targets.

## 4. How to make nut milk

1. **Prepare your ingredients.** Use soaked nuts up to 30 mm, oats, grains or small seeds. Follow normal food-safety rules when you soak and store them.
2. **Add ingredients and clean water to the jug.** Stay within the tested working range and below the fill line. The design target is 0.5–1.5 L (minimum 0.3 L), but please do not assume the prototype is safe at 1.5 L until that has been tested.
3. **Fit the blade assembly, jug and lid.** Check that the fasteners, seals and lid are seated and that the lid/jug switch is engaged.
4. **Choose a program.** Press PROGRAM to pick Manual, Recipe 1 or Recipe 2. While the machine is idle you can change the time and speed.
5. **Press START / STOP.** The motor pulses by itself while the heater brings the water to 100 °C and then keeps it at about 80 °C.
6. **Wait for the end.** After the buzzer, let the blades stop completely and let the milk cool to a safe temperature.
7. **Unplug, then pour.** Pour carefully using the handle. If your assembly has no built-in filter, strain the milk through a food-safe strainer.

## 5. Cleaning and care

1. **Stop, unplug and let everything cool** completely before you take anything apart.

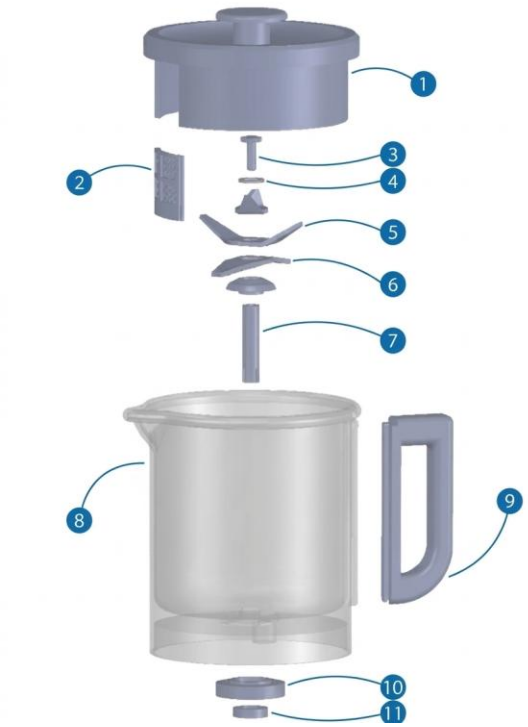
2. **Lift off the jug** and empty out any liquid left inside. Remove the food-contact blade parts only with the approved service procedure.
3. **Wash** the jug, lid and other food-contact surfaces you can reach with mild detergent and a soft brush. Take care around the sharp blades and the seals.
4. **Wipe the base** with a lightly damp cloth, then dry it. Never put the motor, heater wiring, electronics or base in water.
5. **Check for wear.** Look at the shaft seal, coupling, bearings and fasteners for wear, looseness or leaks. Put everything back together only when it is completely dry.

> **CLEANING REMINDER**
> After every 10 hours of motor running time, the display shows **REQCLEAN**. Run the Eco-Self Clean program (see 10.3.4).

## 6. Assembly and parts

The numbers in the drawings match the item numbers in the tables. Use them to find a part when you clean, repair or reassemble the machine.

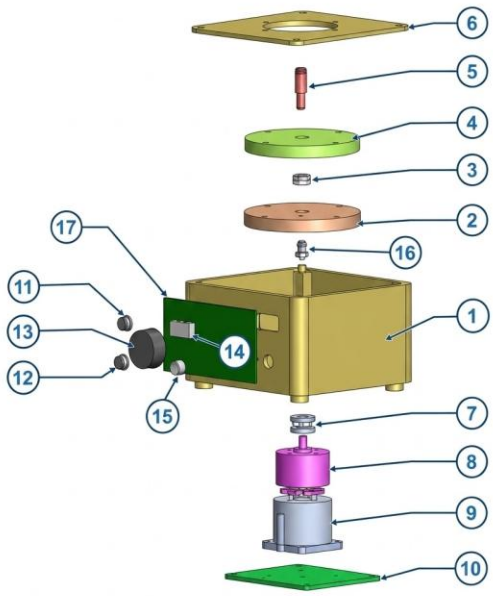
### Jug and blade assembly




*Exploded view: jug and blade assembly*

| Item | Part | Qty |
|---|---|---|
| 1 | Lid | 1 |
| 2 | Filter | 1 |
| 3 | Bolt / screw | 1 |
| 4 | Washer | 1 |
| 5 | Crushing blades | 1 |
| 6 | Shaft holder | 1 |
| 7 | Shaft | 1 |
| 8 | Jug body | 1 |
| 9 | Handle | 1 |
| 10 | Housing | 1 |
| 11 | Bearing | 1 |

### Main base




*Exploded view: motor base, controls and structure*

| Item | Part | Qty |
|---|---|---|
| 1 | Main housing | 1 |
| 2 | Heat insulation plate | 1 |
| 3 | Ball bearing | 1 |
| 4 | Heating plate | 1 |
| 5 | Drive shaft | 1 |
| 6 | Top cover plate | 1 |
| 7 | Motor couplings | 2 |
| 8 | Electric motor / drive unit | 1 |
| 9 | Motor housing / support | 1 |
| 10 | Bottom mounting plate | 1 |
| 11 | Program button | 1 |
| 12 | Speed button | 1 |
| 13 | Time knob | 1 |
| 14 | Display | 1 |
| 15 | Start / Stop button | 1 |
| 16 | Heat sensor | 1 |
| 17 | Arduino board | 1 |

## 7. Troubleshooting

If something goes wrong, stop first, then look for the symptom below.

| What you see | What to check and do |
|---|---|
| **The machine will not start** | Check the mains supply, the seating of the jug and lid, the START / STOP control and the lid switch. Never bypass the switch. |
| **The motor stops or stalls** | Stop and unplug. When the blades have stopped and the machine has cooled, look for an overload or a jam. |
| **Unexpected heating** | Stop and unplug. Have the sensor and heater relay inspected before you use the machine again. |

| What you see | What to check and do |
|---|---|
| **Leak under the jug** | Stop at once. Inspect the shaft seal and the jug for damage. Do not restart until it is repaired. |
| **Strong vibration** | Stop and unplug. Check the blade fixing, coupling, shaft alignment and bearings for wear. |
| **Display or buttons not working** | Disconnect the power. A technician should check the controller wiring and the 5 V supply. See also the blank display in 10.2. |

## 8. Technical specifications

Most values are design targets. The last column shows how far each one has been confirmed.

| Parameter | Value | Status |
|---|---|---|
| **Input supply** | 210–240 V AC, 50/60 Hz | Design requirement |
| **Motor supply** | 12 V DC baseline (RS-997 motor) | Specified |
| **Motor speed** | 6,000–24,000 rpm | Specified |
| **Jug capacity** | 0.5–1.5 L (minimum 0.3 L) | Specified |
| **Heater** | 1,000 W heating plate | Specified |
| **Size** | 19 × 19 × 36.2 cm | Design |
| **Controller** | ATmega328P (Arduino Nano V3) | Specified |
| **Temperature sensor** | NTC 100K, B = 3950 (pin A0) | Specified |
| **Display** | LCD 1602 I2C, address 0x27 | Specified |
| **Motor PWM** | Pin D9, 20 kHz | Specified |
| **Heater control** | Pin D8, relay or SSR output | Specified |
| **Buzzer** | Pin D10, 5-second end alarm | Specified |
| **Controls** | START / STOP D2, TIME D3, lid switch D4 (active LOW), PROGRAM D5, SPEED D7 | Specified |

## 9. Repair, circular design and open-source licence

**Built to be easy repaired**

The machine has three modules (motor, blade and jug) that can be taken apart and replaced one at a time. The team's targets are:

- wear parts such as the shaft seal, blades and motor coupling replaced in **under 10 minutes**;
- full disassembly in **15 minutes or less**, with no more than **three standard tools**.

These are targets. The real service times still have to be checked on the prototype.

**At the end of a part's life**

Keep the food-contact parts (jug, lid, filter and blades) separate from the electrical modules (motor, heater and control board). This makes repair and recycling easier and cuts down on waste.

**Open-source licence**

The Nut Milk Maker is open hardware, published under a CERN hardware licence. The project files are collected in the team's GitHub repository: https://github.com/20725038/Nut-milk-maker-machine. They include the CAD files, wiring schematics, firmware source code, G-code, assembly guide, this user guide and the bill of materials. Some of these are still being completed.

In short, you may study, rebuild, change and share the design as the licence allows. Please read the license text in the repository for the exact conditions.

## 10. Self-test and workaround modes

The machine can test itself, and it keeps working in a reduced way when a part fails. When something stops working, the display tells you which module failed. Where possible, the machine then offers a **workaround,** so it is not completely unusable.

The messages and behavior below follow the firmware specification. Please check the exact wording and beep patterns on your final firmware.

> **IMPORTANT**
> A workaround is only a **temporary fix** until the failed part is repaired.
> All the safety rules in Section 1 still apply. No workaround ever switches off the lid protection.

### 10.1 Run the self-test

1. **Stop** any running cycle and **empty the jug.**
2. **Press and hold START / STOP for 5 seconds.**
3. The machine checks the 5 V supply, the available RAM and EEPROM, then the safety switch, temperature sensor, buzzer, heater relay and motor PWM.
4. **Read the display.** It shows **PASS**, or **FAIL** followed by the name of the failed module.
5. If a module fails, find the message in 10.2 and write the module name down for the technician.

### 10.2 What the display shows and what to do

| The display shows | What it means | What to do |
|---|---|---|
| **PASS** | Every module that was checked answered correctly. | Use the machine as normal. |
| **FAIL: NTC** | The temperature sensor is faulty, so the machine cannot measure the temperature of the liquid. | Use timed heating (see 10.3.1). Stay with the machine every time and have the sensor replaced. |
| **FAIL: MOTOR** | The motor driver signal is faulty. | The motor is switched off and the machine works as a kettle only (see 10.3.2). |

| The display shows | What it means | What to do |
|---|---|---|
| **FAIL: SFTYSWTCH** | The lid / jug switch is not reporting “closed”. | Re-seat the jug and lid, then run the self-test again. If it still fails, stop using the machine and have the switch repaired (see 10.3.3). |
| **FAIL: HEATRRELAY** | The controller cannot switch the heater on or off. | There is no workaround. Do not rely on heating. Unplug the machine and have the relay inspected. |
| **FAIL: BUZZER** | The alarm cannot sound. | There is no workaround. You may not hear the 5-second end alarm, so watch the display. |
| **REQCLEAN** | This is a reminder, not a fault. The motor has run for more than 10 hours. | Run the Eco-Self Clean program with clean water in the jug (see 10.3.4). The counter then resets by itself. |
| **Blank or unreadable display** | The LCD has failed. | The buttons still work on their own. See 10.3.5. |

## 10.3 Workaround modes

### 10.3.1 Temperature sensor failure: timed heating (FAIL: NTC)

**How to switch it on:** press PROGRAM until Eco-Self Clean is selected, then hold the TIME control for 5 seconds.

**What the machine does:** it ignores the temperature sensor. It heats for 30 seconds first, then repeats 5 seconds on and 15 seconds off, and shows a warning on the display.

**The catch:** with no temperature reading, the 80 °C target is not controlled.

**What you should do:** stay with the machine, use it only for short, supervised cycles, and have the sensor replaced.

### 10.3.2 Motor failure: kettle-only mode (FAIL: MOTOR)

**How to switch it on:** press PROGRAM until Eco-Self Clean is selected, then hold the SPEED control for 5 seconds.

**What the machine does:** it switches the motor PWM off, and the machine works as a water boiler.

**The catch:** there is no grinding or mixing. Blend your ingredients separately and use the machine only to heat water.

**What you should do:** have the motor driver and motor coupling checked.

#### 10.3.3 Safety switch failure: no workaround (FAIL: SFTYSWTCH)

**When it happens:** the machine will not start even though the jug and lid are seated correctly.

**What you should do:** re-seat the jug and lid and run the self-test again. If the message stays, stop using the machine and have the switch repaired. The lid protection can never be switched off by the user.

#### 10.3.4 Eco-Self Clean (after REQCLEAN)

**How to start:** press PROGRAM until Eco-Self Clean is selected (the third press from Manual).

**What it does:** it heats to 60 °C and runs the motor at 25% speed in cycles of 10 seconds on and 50 seconds off. It repeats this 10 times (about 10 minutes) and then resets the run-time counter.

#### 10.3.5 Display failure: working without the screen

**What happens:** the LCD does not switch on.

**The catch:** you cannot read the settings or the temperature.

**What you should do:** if you feel comfortable, use the machine as you normally do. The buttons and the machine itself still work without the screen. Have the display repaired when you can.

### 10.4 After a workaround: repair and re-check

1. **Unplug the machine** before you open or change any module.
2. **Replace the failed part** using the repair approach in Section 9.
3. **Run the self-test again** (hold START / STOP for 5 seconds) until the display shows **PASS**.